

MSE211: Computational Method

Lab 3 Report

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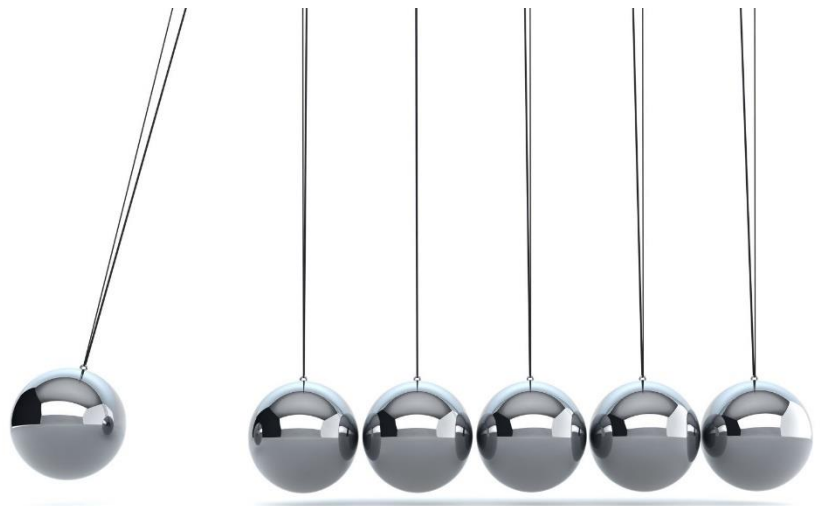


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Abstract

This laboratory study combines numerical and experimental methods to model the motion of a simple pendulum. Using a Linear Servo Base Unit, the pendulum was oscillated experimentally for 8 seconds from five different initial angles. An encoder captured the motion data, which was then analyzed using MATLAB's Simulink. The 'least squares' method was applied to perform a linear regression on the data, estimating the slope of velocity over time as 0.1699 rad/s. This value, alongside four initial angles, was used to predict the maximum linear velocity for the fifth angle, resulting in a velocity of 0.415 m/s for a 0.3365 m long pendulum arm at approximately 72 degrees. The prediction showed a 2.80% discrepancy from the actual data, indicating a successful application of the regression method.

Further analysis explored the pendulum's linear velocity concerning its angle using MATLAB's polyfit tool, revealing a linear relationship between the initial angle and maximum velocity. The study also delved into the pendulum's oscillation modeling. By conducting a force equilibrium analysis in a polar system and applying Newton's second law for rotational systems, a second-order differential equation was derived. This equation was transformed into a system of first-order linear ODEs, which was then solved using the Runge-Kutta 4th order method. This approach, known for its high accuracy in initial value problem solutions, was validated in this study.

MATLAB-generated graphs illustrated the relationship between the initial angles and their corresponding linear velocities, further supporting the linear relationship between initial angle and velocity. While the findings confirm a relatively straightforward relationship, the study successfully demonstrates the effectiveness of computational methods such as linear regression and the Runge-Kutta algorithm in modeling and analyzing pendulum motion.

Introduction

This report explores the dynamics of a pendulum's oscillations, which are governed by a second-order ordinary differential equation indicative of simple harmonic motion. Our objective is to understand and replicate the behavior of this system through both theoretical and computational strategies, utilizing MATLAB for simulation purposes. Initially, the focus is on employing curve fitting techniques to examine the relationship between the pendulum's starting angle and its peak velocity, using data acquired from a laboratory apparatus. We will employ linear regression methods to investigate this relationship and assess the precision of these models by comparing predicted velocities with actual measurements at the maximum angle.

Following the analysis of curve fitting, the report transitions into the modeling phase of the pendulum's motion. Here, we employ the Runge-Kutta 4th-order method, a renowned algorithm for its accuracy in solving initial value problems, to refine our model's predictive capability. The ensuing results, including various graphical representations, will be elaborated upon in the corresponding sections of this report, particularly in the results section, where the findings from both the curve fitting and modeling phases will be discussed in detail.

Linear Regression

Since $\sum a_0 = na_0$, the previous equations can be re-write as

$$0 = \sum y_i - na_0 - \sum a_1 x_i \quad 0 = \sum y_i x_i - \sum a_0 x_i - \sum a_1 x_i^2$$

These are called normal equations and can be solved simultaneously,

$$a_1 = \frac{n \sum x_i y_i - \sum x_i \sum y_i}{n \sum x_i^2 - (\sum x_i)^2}$$

$$a_0 = \bar{y} - a_1 \bar{x}$$

where $\bar{x}$ and $\bar{y}$ are the mean of x and y , respectively.

	Theta 1(11)	Theta 2(21.5)	Theta 3 (35)	Theta 4(52.5)		n	y	x	x^2	x*y		denom	num	A1	A0	
Max Linear Velocity	90	180	277	409		4		90	12	144	1080		3296	27128	8.230583	-7.91748
								180	22	484	3960					
								277	36	1296	9972					
								409	50	2500	20450					
					sum			956	120	4424	35462	y = 0.125x + 0.205				
					mean			239	30							

Utilizing the provided data, we can calculate the maximum velocity for each initial angle (θ). By performing linear regression, we establish the relationship between the initial angle and the maximum velocity, represented by the equation:

$$y = 0.205 + 0.125x$$

Here, y denotes the maximum linear velocity, and x represents the initial angle θ .

Applying this linear regression formula with x set to 70° , we deduce that the maximum linear velocity (y) is 8.955.

Subsequently, we employ MATLAB's polyfit function to conduct a polynomial regression analysis, enabling us to model the maximum linear velocity as a function of the angle. The resulting equation is:

$$V_{\max} = -0.3\theta^2 + 15.1\theta - 84.7$$

Or, in terms of x and y :

$$y = -0.3x^2 + 15.1x - 84.7$$

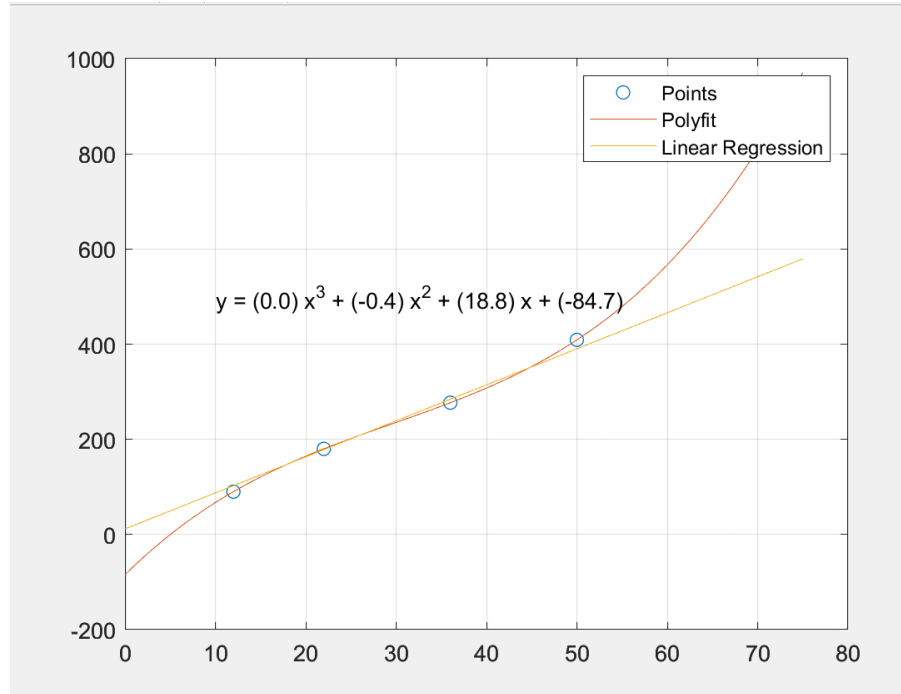
This equation suggests that the maximum linear velocity varies quadratically with the angle.

In our analysis, comparing the two regression methods for an angle of 70 degrees, the polynomial regression predicts a maximum velocity of 9.677, which significantly deviates from the experimental value of 8.554, resulting in a 29.7% error. Conversely, the linear regression provides a closer estimate of 8.955 with an 8.7% error. These findings indicate that the linear model aligns more closely with the experimental data than the polynomial model, particularly at higher angles.

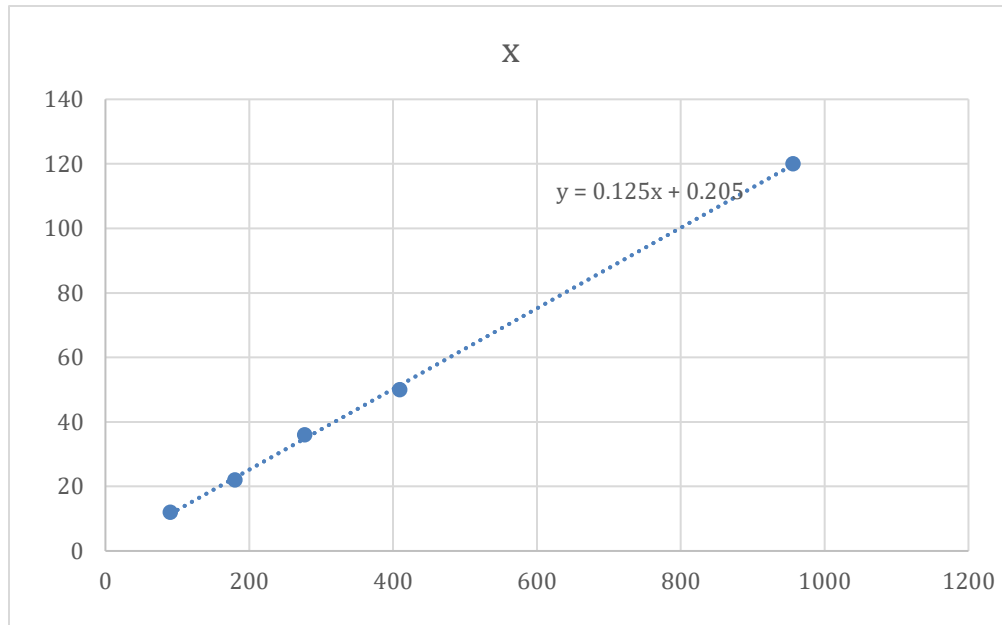
Below is a summary of the results for an angle of 70 degrees:

- **Experimental:** 8.554
- **Linear Regression:** 8.955 (True error: 8.7%)
- **Curve-Fitting (Polynomial):** 9.677 (True error: 29.7%)

This comparison underscores that for the maximum velocity $V_{\max}$ as a function of θ , a linear relationship is more representative than a polynomial one, particularly at higher angles.



The original second-order differential equation, as presented in Equation 1, is decomposed into a system of two first-order ordinary differential equations (ODEs): $\theta' = \omega$ and $\omega' = -(mgr\sin(\theta))/(I + mr^2)$. This transformation facilitates the application of numerical methods for solving the equations.



In our MATLAB implementation, we utilized the Runge-Kutta 4th-order method, a reliable algorithm for approximating the solutions to initial value problems, to solve these ODEs. The details of this algorithm are provided in the appendix for reference.

The outcomes of our simulation, specifically the velocity profiles for all five initial angles (θ), are illustrated in Figure below. These plots provide a visual representation of how the pendulum's velocity evolves over time for different starting angles.

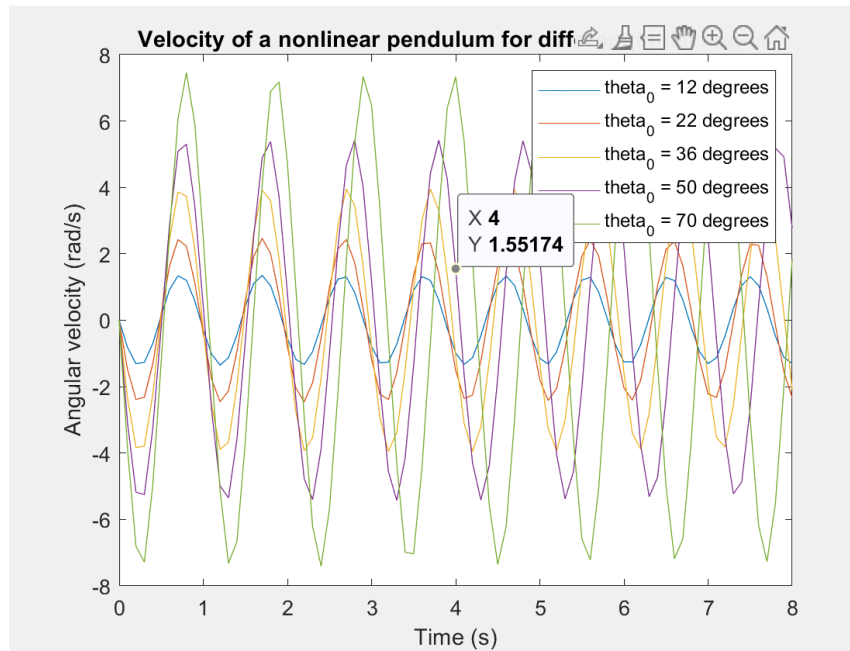


Figure 8: Velocity of a non-linear pendulum for different initial angles

It can be observed that for higher θ_0 values, the angular displacement from equilibrium position is higher and even the max velocity is higher.